Whitepaper: Mycelial Resonance v0.3: The Free-Tier Agentic Swarm

Author: Manus AI

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Abstract

This whitepaper presents Mycelial Echo Forge v0.3, a significant evolution of our biologically inspired AI architecture. Building upon the decentralized and self-healing principles of fungal mycelia, v0.3 integrates enhanced scalability solutions (conceptually DGL/PyTorch Geometric), Bayesian Fusion for improved uncertainty assessment, gradient-based knowledge diffusion, and a revolutionary **Free-LLM-Proxy integration**. The latter enables the use of external, free Large Language Models (LLMs) such as Grok, ChatGPT, Gemini, and Claude without API keys, by leveraging browser automation via Playwright. This transforms Mycelial Echo Forge into a **Free-Tier Agentic Swarm**, empowering every user to operate a powerful, collectively intelligent system using their own free LLM accounts. Additionally, mechanisms for bias/toxicity detection and robust metric logging are introduced to ensure the stability and effectiveness of the swarm.

1. Introduction: From Theoretical Mycelium to Living Swarm

The vision for Mycelial Echo Forge has always been to create an AI architecture that emulates the resilience and decentralized intelligence of fungal mycelia. With version 0.3, we take a decisive step from a theoretical model to a practically applicable, **agentic swarm**. The challenges of scalability, uncertainty assessment, and dynamic knowledge diffusion are addressed through new algorithms. However, the biggest breakthrough is the **Free-LLM-Proxy integration**, which makes state-of-the-art LLMs accessible to everyone, without the need for expensive API keys or complex

authentication procedures. This democratizes access to powerful agentic AI and enables widespread adoption of the mycelial paradigm.

2. Architectural Enhancements in v0.3

Mycelial Echo Forge v0.3 introduces several critical improvements to enhance the robustness, intelligence, and accessibility of the system:

2.1 Scalability: From NetworkX to GNN Frameworks (Conceptual)

While previous versions utilized NetworkX for graph representation, for handling 10,000+ nodes and leveraging GPU acceleration, a transition to specialized Graph Neural Network (GNN) frameworks like **DGL (Deep Graph Library)** or **PyTorch Geometric** is essential. In v0.3, this is conceptually considered by preparing the architecture for seamless integration. These frameworks offer optimized data structures and algorithms for operations on large graphs, significantly boosting the performance of knowledge diffusion and consensus building.

2.2 Uncertainty Assessment: Bayesian Fusion in NTF

Knowledge representation and fusion are refined through the introduction of **Bayesian Fusion** within the NeuroToken Framework (NTF) and Perspective-Driven Consensus (PDP). Instead of simple averaging, uncertainties and confidence values of agents are now incorporated into consensus building. Agents whose knowledge falls below a confidence threshold of `<0.4` are temporarily isolated to safeguard the quality of collective knowledge and prevent the spread of erroneous or uncertain information. This leads to a more robust and trustworthy consensus.

2.3 Dynamic Knowledge Diffusion: Gradient-Style with Decay

The previous simple 0.7/0.3 mixing for knowledge diffusion is replaced by a **gradient-based approach with decay and direction**. This more accurately emulates biological diffusion, where knowledge is not just shared but actively flows towards “knowledge gradients” and loses influence over distance. This enables more efficient and targeted distribution of relevant information within the mycelial network and promotes the emergence of specialized knowledge clusters.

2.4 Free-LLM-Proxy: External LLM Integration without API Keys

This is the core of v0.3. The **Free-LLM-Proxy** enables the integration of external online LLMs (Grok, ChatGPT, Gemini, Claude) using their free tiers, **without requiring API keys**. The solution is based on:

- **Browser Automation (Playwright):** A local Playwright script launches a browser where the user manually logs into their preferred LLMs (including OAuth and 2FA).
- **Webhook Communication:** The mycelium sends tasks as webhook requests to the locally running proxy.
- **Session Utilization:** The proxy uses the active browser sessions to send requests to the LLMs and extract responses.
- **Local Processing:** All interactions remain local to the user's system, ensuring maximum security and privacy.

This feature democratizes access to powerful AI and allows anyone to operate a highly intelligent agentic system by utilizing their existing free-tier accounts. For subscription users with API access, the proxy will also serve as a central interface to leverage multiple online LLMs in alignment with mycelial progress.

2.5 Bias and Toxicity Detection

To protect the integrity of the mycelial network, an **auto-detector for bias and toxicity** is implemented. Based on simple rules (conceptually expandable by APIs like Perspective API), toxic content is identified. Nodes that generate or spread such content can be isolated or their contributions to consensus building reduced, to prevent the spread of harmful information and foster a healthy, ethical mycelium.

2.6 Logging and Metrics: MLflow and Prometheus-Style

For comprehensive monitoring and analysis of mycelial dynamics, **MLflow** and **Prometheus-style metrics** are integrated. Key metrics such as `Drift-Rate` (change in collective knowledge over time), `Heal-Time` (time to recover from disturbances), and `Consensus-Trend` (evolution of consensus quality) are recorded. This provides detailed insights into the swarm's behavior, parameter optimization, and early detection of anomalies.

3. Mathematical Foundations and Algorithms (v0.3)

3.1 Bayesian Fusion for Consensus

Consensus formation C is now formulated as a weighted average of agent knowledge vectors k_i considering their confidence conf_i:

$$C = \frac{\sum_{i=1}^{N} \text{conf}_i \cdot k_i}{\sum_{i=1}^{N} \text{conf}_i}$$

Here, conf_i is derived from the agent's perspective and its history. Agents with $\text{conf}_i < 0.4$ are excluded from the consensus calculation to enhance robustness.

3.2 Gradient-Based Knowledge Diffusion

Knowledge diffusion for an agent a_i is updated by:

$$k_i^{(t+1)} = \text{normalize}(k_i^{(t)} + \eta \sum_{j \in N(i)} w_{ij} \cdot (k_j^{(t)} - k_i^{(t)}) \cdot e^{-\lambda d_{ij}})$$

where η is the learning rate, w_{ij} is the edge weight, $(k_j^{(t)} - k_i^{(t)})$ is the knowledge gradient, d_{ij} is the distance (e.g., number of hops) between a_i and a_j, and $e^{-\lambda d_{ij}}$ is an exponential decay factor. `normalize` ensures NTF normalization.

4. Conclusion and Outlook

Mycelial Echo Forge v0.3 represents a turning point in the development of agentic AI. By combining biologically inspired architectures with innovative solutions for scalability, uncertainty assessment, and, most importantly, the **democratic integration of Free-Tier LLMs**, we create a system that is not only powerful and resilient but also accessible to everyone. The vision of a "Free-Tier Agentic Swarm" becomes a reality, and we invite the global community to be part of this revolution.

Future versions will focus on the full implementation of DGL/PyTorch Geometric, the development of a Mobile App Wrapper (v0.4), and the creation of Self-Hosted Swarm functionality (v0.5) to further decentralize and strengthen the mycelial network.

References

[1] IrsanAI GitHub Repository: https://github.com/IrsanAI/mycelial-echo-forge [2] NeuroToken Framework (NTF) Concept: (Reference to be added after NTF whitepaper publication) [3] Perspective-Driven / Multi-Model-Consensus (PDP) Concept: (Reference to be added after PDP whitepaper publication) [4] Playwright Documentation: https://playwright.dev/ [5] FastAPI Documentation: https://fastapi.tiangolo.com/

This whitepaper is a draft. For a final version, review by Claude.ai is recommended to ensure corrections and improvements.